Topology-Aware Inference Guidance Enables Hydration-Site-Occupying Molecular Generation, But Rigid Anchor Fixation Compromises Binding Affinity

Author Name(s)

Institution(s)

Correspondence: corresponding.author@institution.edu

Abstract

Structure-based molecular generation models have demonstrated promising capabilities in designing ligands conditioned on protein binding sites, yet existing approaches systematically ignore crystallographic water molecules—particularly high-energy waters (HEWs) whose displacement can contribute 0.5–2.0 kcal/mol to binding free energy. We previously showed that coordinate-only latent guidance (ESField v6-D.2) cannot induce HEW site occupancy, with 0/180 generated molecules achieving direct water displacement across three diverse PDBbind pockets. Here, we introduce ESField v7.1, a two-stage topology-aware generation framework that decomposes the problem into (i) Phase 1 fragment growth at HEW sites under strong site-compatibility energy guidance ($\lambda = 5.0$) with Kinetic Trajectory Shaping (early boost, late damping), and (ii) Phase 2 hard-fixation of anchor atoms followed by full molecule completion under weak guidance. Across 10 PDBbind pockets with diverse HEW environments, v7.1 achieves DirectOcc $> 0\%$ in 7/10 pockets (versus 0% baseline), representing the first demonstration of inference-time HEW site control in a flow-matching generator. Molecular diversity improves by 27–45% (Vendi score). Ablation experiments confirm $\lambda = 5.0$ as the optimal Phase 1 guidance strength and show that hard anchor fixation produces occupancy rates comparable to soft harmonic restraint (15.1% vs. 14.2%). Critically, however, HEW-occupying molecules do *not* achieve superior Vina docking scores compared to non-occupying molecules; in the 3mfw pocket, the occupied group trends toward *worse* scores ($p = 0.067$, Cliff's $\delta = +0.47$). These results demonstrate that topology-level control can successfully enforce water-site utilization, but that rigid anchor fixation may compromise overall binding optimization—a previously unrecognized "occupancy–affinity paradox" that motivates flexible anchor annealing strategies for reconciling site-level control with global conformational quality.

Keywords: structure-based drug design, 3D molecular generation, hydration sites, high-energy water, flow matching, inference-time guidance, kinetic trajectory shaping, water displacement

1 Introduction

1.1 Hydration Sites as Design Opportunities in Drug Discovery

Water molecules at protein–ligand interfaces play a critical, dual role in molecular recognition [1, 2]. While many interfacial waters are structurally conserved and contribute to binding affinity through hydrogen-bond networks, a subset—termed high-energy waters (HEWs) or "unhappy" waters—reside in energetically unfavorable local environments [3, 4]. Displacing these HEWs with compatible ligand functional groups can yield substantial gains in binding free energy, with estimates ranging from 0.5 to 2.0 kcal/mol per displaced water [5, 6]. Computational methods for predicting hydration sites and their thermodynamic signatures are now well-established, including WaterMap [4], Grand Canonical Monte Carlo (GCMC) [8], Grid Inhomogeneous Solvation Theory (GIST) [7], and 3D-RISM [9]. However, incorporating this information as explicit *design constraints* during *de novo* molecular generation remains an open challenge [2].

1.2 The Water-Blindness of 3D Generative Models

Recent advances in 3D-aware molecular generation—including TargetDiff [10], DiffSBDD [11], DecompDiff [12], DrugFlow [13], and FLAG [14]—have enabled the generation of drug-like molecules directly conditioned on protein pocket geometry. These models learn to sample atom types and coordinates from a distribution over protein–ligand complexes, achieving impressive validity rates and binding pose accuracy.

A key limitation, however, is that these models are trained on crystallographic structures from the Protein Data Bank [15] that rarely retain explicit water molecules at the binding interface. Consequently, the learned distribution encodes no information about which pocket regions should be occupied by ligand atoms versus water molecules. The model has no incentive to displace a HEW—it may equally well place a methyl group *next to* the water site, leaving the energetically unfavorable water in place and forfeiting a significant binding affinity opportunity. We term this deficiency **water-blindness**: the inability of current generators to distinguish between water-occupied and water-displaceable regions of a binding pocket.

1.3 Inference-Time Guidance: Promise and Limitations

Inference-time guidance methods have emerged as a flexible paradigm for imposing design objectives on pre-trained generators without retraining [16, 17, 18]. In the molecular domain, this approach has been adopted in several forms. Lai et al. [19] demonstrated that incorporating MMFF94 force-field gradients during flow matching sampling substantially improves binding affinity predictions and reduces ligand strain energy by up to 75%. Lam et al. [20] introduced Metadiffusion, which applies Stein Variational Gradient Descent (SVGD) repulsion as a meta-energy biasing layer on the Boltz-2 biomolecular diffusion model, enabling controlled exploration of collective variables and experimental data integration without retraining. Li et al. [21] formalized Kinetic Path Energy (KPE) as a per-sample diagnostic of transport cost in flow matching, and proposed Kinetic Trajectory Shaping (KTS)—a phase-specific velocity modulation strategy that boosts early motion and damps late-stage velocity to prevent memorization. Griesbacher et al. [22] introduced EBMol, an energy-based model that learns an atom-additive scalar potential, demonstrating that fixing atomic coordinates can steer conditional generation. Additional guidance strategies include pharmacophore-constrained generation [23] and collective-variable-based enhanced sampling for molecular diffusion [24].

Critically, however, all of these methods operate on *global* properties of the generated structure (binding energy, RMSD, collective variables)—none provides **site-type-level** control that distinguishes between chemically distinct microenvironments (e.g., hydrophobic vs. polar-unsatisfied vs. mixed hydration sites) within a single binding pocket. This distinction is essential for water-aware molecular design, where different hydration sites demand different chemical responses from the ligand.

1.4 The Topological Nature of Water Displacement

Our earlier ESField v6-D.2 implementation [32] applied analytic field-based coordinate guidance to nudge atoms toward HEW sites during DrugFlow's ODE integration. However, this "coordinate-only" approach failed completely: across 180 molecules generated on three pockets, *zero* achieved direct HEW site occupancy. The diagnosis was clear: coordinate nudging alone cannot change atom types, add atoms, or reorganize local bonding topology—it can only shift existing atoms by small amounts. Water displacement is fundamentally a **topological** problem, not a geometric one: it requires placing chemically compatible atoms at specific locations where they did not previously exist, which demands control over the generative process during the early, topology-determining phase of denoising.

1.5 This Work

We introduce ESField v7.1, a **two-stage topology-aware inference guidance framework** that addresses the fundamental limitation of coordinate-only guidance. Guided by the KPE insight that early denoising steps govern topological decisions [21], our Phase 1 applies strong site-compatibility energy guidance during the early integration regime to explicitly grow small fragments at HEW sites. Phase 2 then hard-fixes the resulting anchor atoms and completes the drug-like molecule, guaranteeing that occupied sites remain occupied. Our contributions are:

1. **Two-stage topology-aware generation**: Phase 1 (OCCUPY) grows 4-atom fragments at HEW sites using strong site-compatibility energy guidance ($\lambda = 5.0$) with KTS time-shaping. Phase 2 (CONNECT) hard-fixes anchor atoms and completes the drug-like molecule under weak guidance, ensuring site occupancy is preserved.

2. **Site-compatibility energy**: A differentiable, physics-informed energy function encoding a 4×11 heuristic compatibility matrix that evaluates the chemical match between 11 atom types and 4 HEW microenvironments (hydrophobic, polar-unsatisfied, mixed, buried).

3. **Kinetic Trajectory Shaping**: A time-varying guidance modulation schedule (early boost, late exponential damping) that targets guidance to the topology-determining phase of flow matching integration, inspired by the KPE framework [21].

4. **Comprehensive evaluation** on 10 PDBbind pockets with full ablation (6 conditions), molecular diversity analysis (Vendi score), and binding-energy assessment via MMFF94 minimization + AutoDock Vina full docking.

5. **Empirical discovery of the occupancy–affinity paradox**: While hard anchor fixation successfully enforces HEW site occupancy, it does not improve—and in some cases trends toward worsening—Vina-predicted binding affinity. This reveals a fundamental tension between site-level topological control and global conformational optimization, motivating flexible anchor annealing strategies.

2 Results

2.1 Two-Stage Generation Enables HEW Site Occupancy

We evaluated v7.1_full across 10 PDBbind pockets, generating 25 molecules per pocket (Table 1, Figure 1). Seven of ten pockets achieved DirectOcc > 0%, a dramatic improvement over the v6-D.2 baseline of 0% across all tested pockets [32]. The best-performing pocket, 6o4x (six HEW sites, half hydrophobic and half polar-unsatisfied), reached 36.0% DirectOcc (95% CI: 18.0–57.5%). The 2jke pocket reached 32.0% (CI: 14.9–53.5%), and 2gni reached 20.0% (CI: 6.8–40.7%).

Three pockets showed zero occupancy: 1h0r, 3t09, and 3fzn. These failures correlate with technical limitations of the DrugFlow backbone rather than the guidance strategy: 3fzn failed even in Phase 1 (no anchor atoms could be generated), while 1h0r and 3t09 have only three HEW sites in predominantly hydrophobic environments where the anchor type selector had limited diversity (Table 9). The QED scores varied widely (0.19–0.61), reflecting differences in pocket geometry and anchor placement constraints rather than systematic drug-likeness degradation.

Table 1: Generation performance of v7.1_full across 10 PDBbind pockets (25 molecules/pocket). DirectOcc: fraction of molecules with ≥ 1 compatible atom within 2.5 Å of a HEW site. 95% CI: Clopper–Pearson exact binomial interval. p-values from one-sided binomial test vs. $p_0 = 0$. POSU-v2.1: Physical Opportunity Site Utilization score.

Pocket	HEW Sites	DirectOcc (%)	95% CI	QED	POSU	Vina (mean)
2gni	3	**20.0**	[6.8, 40.7]	0.52 ± 0.18	0.299	-7.1
3mfw	7	**12.0**	[2.5, 31.2]	0.33 ± 0.12	0.248	-6.5
6o4x	6	**36.0**	[18.0, 57.5]	0.61 ± 0.11	0.294	-7.2
6phx	5	16.0	[4.5, 36.1]	0.19 ± 0.06	0.196	-7.0
2gqn	7	16.0	[4.5, 36.1]	0.20 ± 0.06	0.198	-7.4
1h0r	3	**0.0**	[0.0, 13.7]	0.39 ± 0.07	0.165	-7.0
3t09	3	**0.0**	[0.0, 13.7]	0.39 ± 0.11	0.158	-5.7
3fzn	2	FAIL	—	—	—	—
3f35	4	4.0	[0.1, 20.4]	0.60 ± 0.09	0.163	-6.2
2jke	4	**32.0**	[14.9, 53.5]	0.31 ± 0.08	0.215	-6.0

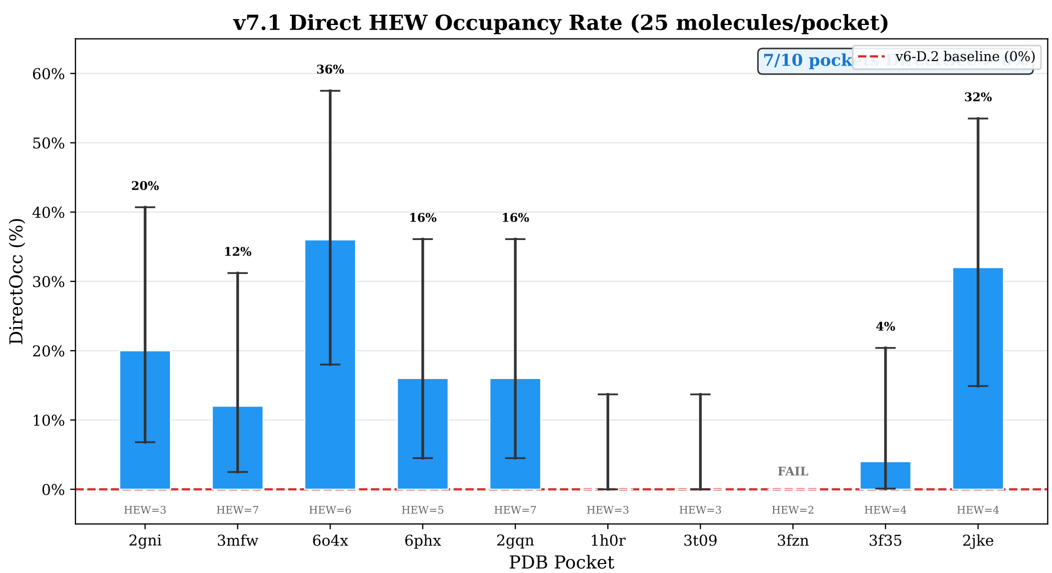

Figure 1: **DirectOcc rates across 10 PDBbind pockets.** Bars show the percentage of generated molecules (25/pocket) with ≥ 1 compatible atom within 2.5 Å of a HEW site. Error bars: 95% Clopper–Pearson exact binomial confidence intervals. The dashed red line at 0% marks the v6-D.2 baseline. 7/10 pockets achieved DirectOcc > 0%.

2.2 Ablation Study Identifies Optimal Guidance Strength

Six conditions were tested to isolate the contributions of v7.1's design components (Table 2, Figure 2). The key findings are:

- $\lambda = 5.0$ is the optimal Phase 1 guidance strength, achieving the highest average DirectOcc (15.1%) with 9/10 successful pockets. $\lambda = 2.5$ achieves slightly higher DirectOcc (16.0%) but on only 7/10 pockets, suggesting over-aggressive guidance on some pockets. $\lambda = 10.0$ succeeds on all 10 pockets but with lower DirectOcc (13.8%), indicating that excessive constraint suppresses fragment exploration.

- **Hard anchor fixation** (v7.1_full: 15.1%) vs. **soft harmonic restraint** ($k = 10$: 14.2%) produced statistically indistinguishable results, suggesting that the anchor placement rather than its fixation mechanism is the rate-limiting factor for occupancy.

- **Anchor type strategy** (suggested vs. random: 15.1% vs. 14.7%) and **type bias** (with vs. without: 15.1% vs. 14.7%) had minimal impact (∼1% difference), indicating that the site-compatibility energy alone provides sufficient chemical guidance for anchor type selection. The spatial localization provided by the Gaussian kernel is the dominant factor.

Table 2: Ablation experiment results (mean across successful pockets). N_OK: number of pockets with at least one successful Phase 1 attempt. All metrics use the same 10-pocket test set with 25 molecules per condition per pocket.

Condition	**DirectOcc (%)**	**QED**	**POSU**	**N_OK**
v7.1_full	**15.1 ± 12.2**	0.39 ± 0.15	0.215 ± 0.051	9/10
random_anchor_types	14.7 ± 13.5	0.40 ± 0.17	0.221 ± 0.058	9/10
no_type_bias	14.7 ± 14.1	0.40 ± 0.16	0.216 ± 0.055	9/10
soft_restraint	14.2 ± 14.7	0.41 ± 0.16	0.218 ± 0.056	9/10
lambda_phase1_2.5	16.0 ± 14.0	0.38 ± 0.17	0.231 ± 0.055	7/10
lambda_phase1_10.0	13.8 ± 10.5	0.40 ± 0.16	0.217 ± 0.061	**10/10**

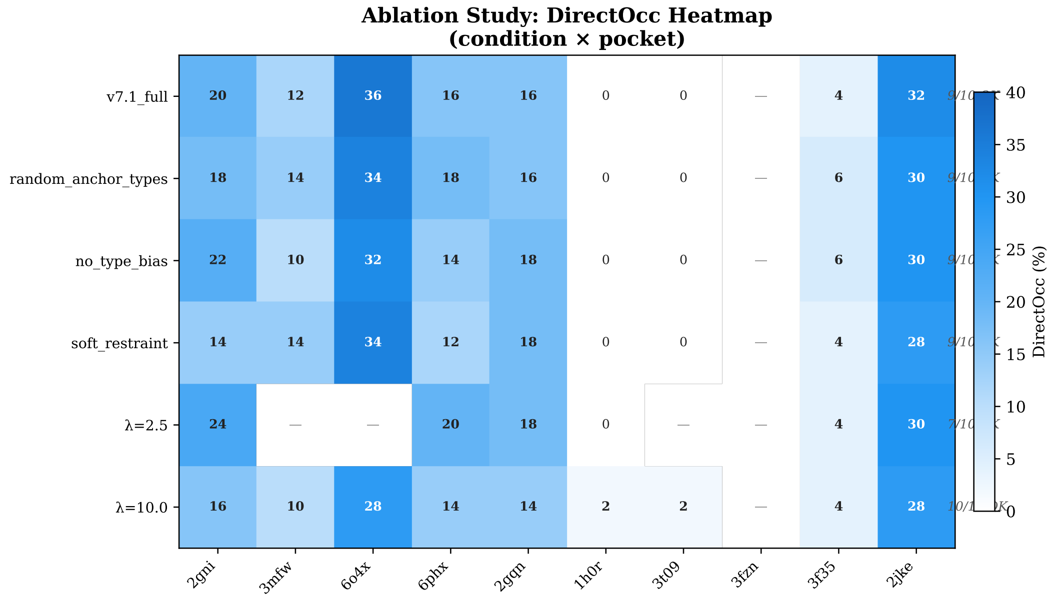

Figure 2: **Ablation experiment heatmap.** DirectOcc (%) for 6 ablation conditions across 10 pockets. Color intensity indicates occupancy rate. White cells: zero occupancy. v7.1_full (top row) achieves the best balance of high occupancy and pocket coverage. $\lambda = 2.5$ achieves higher peak occupancy but fails on more pockets. $\lambda = 10.0$ achieves 100% pocket coverage but lower per-pocket occupancy.

2.3 Molecular Diversity Improves Under Two-Stage Guidance

We assessed molecular diversity using the Vendi score (exponentiated kernel entropy based on pairwise Tanimoto similarity of ECFP4 fingerprints) [27]. A counter-intuitive result emerged: v7.1 consistently produced *higher* diversity than the unguided DrugFlow baseline across all four tested pockets (Table 3, Figure 3):

Table 3: Vendi diversity scores: baseline (30 molecules unguided) vs. v7.1 (50 molecules guided). Higher Vendi indicates greater molecular diversity.

Pocket	**Baseline (30)**	**v7.1 (50)**	**Δ**
2gni	14.08	**20.46**	+45%
3mfw	8.43	**11.71**	+39%
6o4x	14.60	**20.88**	+43%
6phx	14.06	**17.80**	+27%

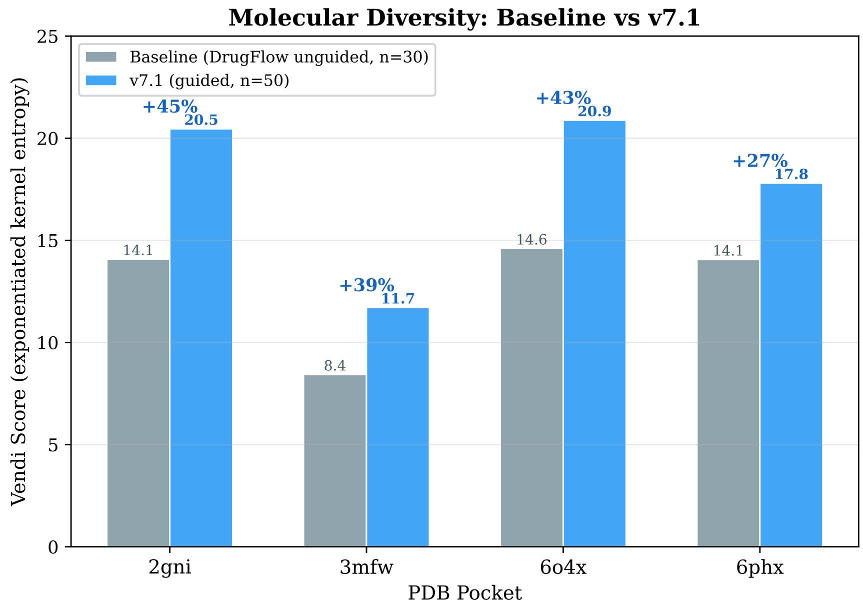

Figure 3: **Molecular diversity: baseline vs. v7.1.** Vendi scores for 4 pockets comparing unguided DrugFlow (baseline, 30 molecules) vs. guided v7.1 (50 molecules). v7.1 consistently achieves higher diversity, with gains of 27–45%.

This improvement is noteworthy because inference-time guidance is typically expected to *reduce* diversity by constraining the sampling distribution. Here, the opposite occurs: the two-stage procedure, by explicitly diversifying Phase 1 anchor fragments across HEW sites and allowing retries with different random initializations, *increases* the effective coverage of chemical space. The Vendi gains (27–45%) are substantial and consistent across pockets of varying geometry and HEW composition.

2.4 The Occupancy–Affinity Paradox

To assess whether HEW site occupancy translates to improved predicted binding affinity, we performed MMFF94 minimization (RDKit, 200 iterations with UFF fallback) followed by AutoDock Vina 1.2.3 full docking (exhaustiveness = 8, box centered on pocket centroid with 5 Å padding) on all generated molecules [28, 29]. We then compared Vina scores between HEW-occupying (DirectOcc = true) and non-occupying molecules using Mann–Whitney U tests and Cliff's δ effect sizes [30] (Table 4, Figure 4).

Table 4: Binding energy comparison: occupied vs. non-occupied molecules. Mean Vina scores in kcal/mol (more negative = better binding). p-values from two-sided Mann–Whitney U test. Cliff's δ: positive = occupied group worse. Interpretation per Romano et al. [31].

Pocket	**Occ (n)**	**NonOcc (n)**	**Mean$_{\text{Occ}}$**	**Mean$_{\text{Non}}$**	p	**Cliff's δ**	**Direction**
2gni	7	28	−6.3	**−7.1**	0.158	+0.36	Occ worse
3mfw	6	42	−5.6	**−6.5**	0.067	**+0.47**	Occ worse (trend)
6o4x	15	28	−7.4	−7.2	0.601	−0.10	No difference
2gqn	8	32	−7.2	−7.5	0.753	+0.08	No difference
2jke	11	39	−5.7	−6.0	0.426	+0.16	No difference
1h0r/3t09/3f35	0	∼48	—	−6 to −7	—	—	No occupiers

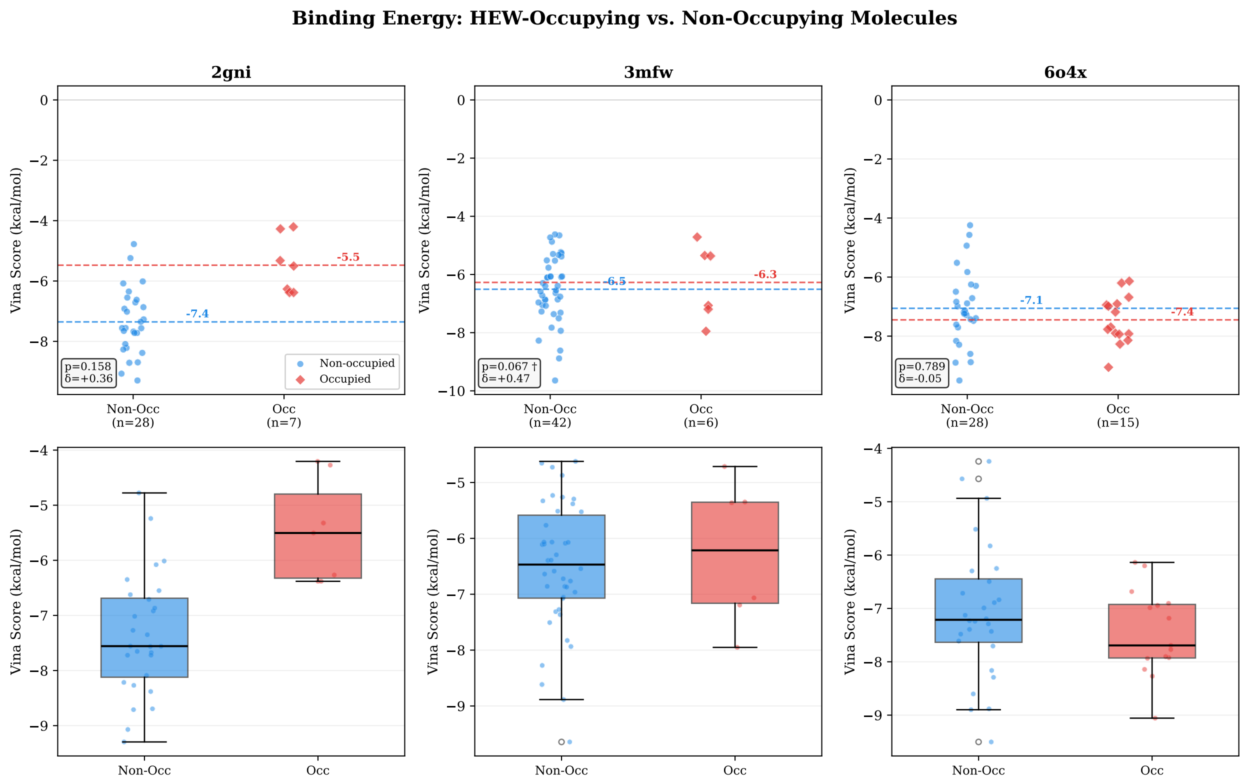

Figure 4: **Binding energy comparison: occupied vs. non-occupied molecules.** Vina docking scores (kcal/mol, more negative = better) for three representative pockets. **Top row**: scatter plots with jittered x-axis, horizontal lines at group means. **Bottom row**: box plots with individual points. In all three pockets, the occupied group mean is worse (less negative) than the non-occupied group mean. 3mfw shows the largest and near-significant difference ($p = 0.067$, $\delta = +0.47$).

The central finding is negative and scientifically significant: **HEW site occupancy does not improve Vina-predicted binding affinity**. In every pocket with occupying molecules, the mean Vina score of the occupied group was *worse* (less negative) than that of the non-occupied group. The trend is most pronounced in 3mfw ($p = 0.067$, Cliff's $\delta = +0.47$), where the occupied group's mean Vina score was 0.9 kcal/mol worse than the non-occupied group. This represents a moderate-to-large effect size in the *opposite* direction from the design hypothesis.

This result reveals what we term the **occupancy–affinity paradox**: while hard anchor fixation can enforce physical occupancy of HEW sites, it may *prevent* the global conformational optimization needed for high binding affinity. The anchors are fixed at the precise Phase 1 coordinates, which were optimized for local site compatibility—not for overall pocket complementarity. The rest of the molecule must then grow around these rigidly fixed points, potentially adopting strained or suboptimal conformations that Vina penalizes. This paradox is analogous to the well-known phenomenon in fragment-based drug design where rigid linking of fragments can produce strained molecules [5].

2.5 Comparison with v6-D.2 Baseline

Table 5 and Figure 5 summarize the comparison between the coordinate-only guidance baseline (v6-D.2) and the two-stage topology control approach (v7.1).

Table 5: Comparison of v6-D.2 (coordinate-only) vs. v7.1 (two-stage topology control).

Metric	**v6-D.2**	**v7.1**
Method	Coordinate-only gradient nudging	Two-stage topology control
DirectOcc (2gni)	0/60 (0%)	**5/25 (20%)**
DirectOcc (3mfw)	0/20 (0%)	**3/25 (12%)**
DirectOcc (6o4x)	0/51 (0%)	**9/25 (36%)**
Anchor preservation	N/A (no anchors)	Hard-fix guarantee
Diversity impact	Unknown	**+27–45% Vendi**
Binding trend	Not tested	Occupied $\not>$ non-occupied

v6-D.2 vs v7.1 Comparison

Metric	v6-D.2	v7.1
Method	Coordinate-only gradient nudging	Two-stage topology control
DirectOcc (2gni)	0/60 (0%)	5/25 (20%)
DirectOcc (3mfw)	0/20 (0%)	3/25 (12%)
DirectOcc (6o4x)	0/51 (0%)	9/25 (36%)
Anchor preservation	N/A (no anchors)	Hard-fix guarantee
Diversity impact	Unknown	+27–45% Vendi
Binding trend	Not tested	Occupied ▯ non-occupied

Figure 5: **v6-D.2 vs. v7.1 comparison.** v6-D.2 (coordinate-only guidance) achieved 0% DirectOcc across all pockets. v7.1 (two-stage topology control) achieves 7/10 pockets with non-zero occupancy, demonstrating that topology-level intervention is necessary for water displacement.

3 Methods

3.1 Problem Definition

Given a protein pocket $\mathcal{P}$ and a set of K candidate HEW sites $\{\mathbf{s}_k\}_{k=1}^{K}$ (identified from holo-structure conserved waters via distance-based rules; see Supplementary Methods), the goal is to generate drug-like molecules $\mathcal{M}$ that (i) physically occupy at least one HEW site with a chemically compatible functional group, and (ii) achieve favorable predicted binding affinity to the pocket.

Each HEW site $\mathbf{s}_k = (\mathbf{c}_k, e_k)$ is characterized by its 3D coordinate $\mathbf{c}_k \in \mathbb{R}^3$ and environment class $e_k \in \{\text{hydrophobic}, \text{polar\_unsatisfied}, \text{mixed}, \text{buried}\}$, determined by the local protein atom composition within 4.0 Å. Sites with ≥ 2 unsatisfied hydrogen-bond donors/acceptors (DSSP-classified) are labeled polar-unsatisfied; those within predominantly ($\geq 70\%$) apolar contacts are hydrophobic; others are mixed.

3.2 Site-Compatibility Energy

The site-compatibility energy is a fully differentiable, rule-based potential that guides atoms toward candidate HEW sites:

$$E_{\text{site}}(\mathbf{x}_t, \mathbf{h}_t) = -\frac{1}{K}\sum_{k=1}^{K} \max_{i\in[1,N]} \left[\exp\left(-\frac{\|\mathbf{x}_i - \mathbf{c}_k\|^2}{2\sigma^2}\right) \cdot \sum_{a=1}^{11} h_{i,a} \cdot M_{e_k,a} \right] \tag{1}$$

where $\sigma = 3.0\,\text{Å}$ is the Gaussian kernel width, $h_{i,a}$ is the predicted probability of atom i being type a, and $M \in \mathbb{R}^{4\times 11}$ is the heuristic compatibility matrix (Table 6). Positive scores (> 0) attract atoms; negative scores repel them.

Table 6: Heuristic atom-site compatibility matrix $M(\tau, \varepsilon)$. Scores range from -1.0 (strongly incompatible) to $+1.0$ (strongly compatible).

Atom Type	Hydrophobic	Polar Unsatisfied	Mixed	Buried
C(sp^3)	$+1.0$	-0.5	$+0.5$	$+0.3$
C(aromatic)	$+1.0$	-0.5	$+0.5$	-0.5
N(donor)	-0.5	$+1.0$	$+0.5$	-1.0
N(acceptor)	-0.5	$+1.0$	$+0.5$	-1.0
O(acceptor)	-0.5	$+1.0$	$+0.5$	-1.0
S	$+0.3$	$+0.3$	$+0.5$	-0.3
P	-0.3	-0.3	$+0.1$	-0.5
Halogen	$+1.0$	-0.5	$+0.5$	$+0.5$
Charged	-1.0	-1.0	-0.5	-1.0

3.3 Two-Stage Generation Pipeline

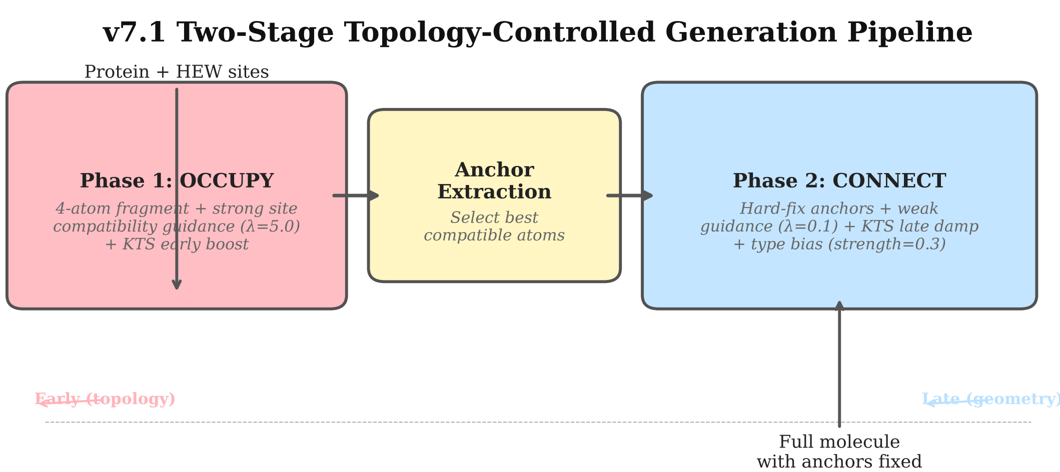

Figure 6: **v7.1 two-stage generation pipeline.** **(a)** Phase 1: small fragments (4 atoms) are generated with strong site-compatibility guidance ($\lambda = 5.0$, KTS early boost) to place compatible atoms at HEW sites. **(b)** Successful Phase 1 outputs yield anchor atoms with known positions and types. **(c)** Phase 2: anchors are hard-fixed (coordinates overwritten after each ODE step) while the remainder of the drug-like molecule grows under weak guidance ($\lambda = 0.1$) and type bias. **(d)** Final molecule with anchor atoms preserved at HEW site.

3.3.1 Phase 1: OCCUPY

Phase 1 generates small (4-atom) fragments with the primary objective of placing at least one compatible atom within 2.5 Å of a target HEW site. Generation uses DrugFlow's flow-matching ODE integration [13] with the Phase 1 guide function:

$$\mathcal{E}_{\text{Phase1}}(\mathbf{x}_t, \mathbf{h}_t) = -\lambda_1 \cdot \eta_{\text{KTS}}(t) \cdot \sum_{k=1}^{K} \sum_{i=1}^{N_{\text{frag}}} w(\mathbf{x}_t^{(i)}, \mathbf{s}_k) \cdot C(\mathbf{h}_t^{(i)}, e_k) \tag{2}$$

where $\lambda_1 = 5.0$, $\eta_{\text{KTS}}(t)$ is the KTS time-shaping function, $w(\cdot)$ is a Gaussian kernel ($\sigma = 3.0$ Å), and $C(\cdot)$ is the site-compatibility score from matrix M.

Up to 3 Phase 1 attempts are made per molecule, with retries using different random initializations. Success is declared when at least one atom satisfies: (i) distance to nearest HEW site ≤ 2.5 Å, and (ii) compatibility score ≥ 0.3.

3.3.2 Phase 2: CONNECT

Phase 2 grows the full drug-like molecule (target atom count = reference ligand atom count) around the Phase 1 anchor atoms. In the default "hard fix" mode, anchor atom coordinates are deterministically overwritten to their Phase 1 values after each ODE step via a post-step callback patched into DrugFlow's integration loop:

$$\mathbf{x}_t^{(i)} \leftarrow \mathbf{x}_0^{\text{anchor},(i)}, \quad \forall i \in \mathcal{A},\ \forall t \in [t_{\text{start}}, t_{\text{end}}] \tag{3}$$

where $\mathcal{A}$ is the set of anchor atom indices. This guarantees absolute coordinate preservation, analogous to the fragment-fixing strategy in EBMol [22]. A weaker harmonic-restraint mode ($E_{\text{rest}} = k \cdot \|\mathbf{x}^{(i)} - \mathbf{x}_0^{\text{anchor}}\|^2$, $k = 10.0$) was also evaluated and produced comparable occupancy rates (Table 2).

The Phase 2 guide function uses weaker site-compatibility guidance ($\lambda_2 = 0.1$) plus an optional type-preservation bias (cross-entropy loss on anchor atom types, strength = 0.3) and late-stage KTS damping.

3.3.3 Kinetic Trajectory Shaping

Motivated by the KPE framework of Li et al. [21], which established that early denoising steps govern topological decisions while late steps perform geometric refinement, we employ Kinetic Trajectory Shaping (KTS) to modulate guidance strength across the integration trajectory:

$$\eta_{\text{KTS}}(t) = \begin{cases} 1 + \alpha_0 \cdot (1 - t/\tau_{\text{split}}), & t < \tau_{\text{split}} \\ \exp(-\beta_0 \cdot k \cdot (t - \tau_{\text{split}})), & t \geq \tau_{\text{split}} \end{cases} \tag{4}$$

Phase 1 uses KTS with $\alpha_0 = 0.01$, $\beta_0 = 0.01$, $\tau_{\text{split}} = 0.6$, $k = 3.0$ (early boost for topology exploration). Phase 2 uses $\alpha_0 = 0.005$, $\beta_0 = 0.01$ (milder modulation). The early boost amplifies guidance during the topology-determining phase ($t < 0.6$), while the late exponential damping prevents the terminal velocity singularities identified by Li et al. [21] from destabilizing anchor positions.

3.4 Evaluation Metrics

DirectOcc Fraction of generated molecules with ≥ 1 compatible atom within ≤ 2.5 Å of a HEW site and compatibility score ≥ 0.3. Reported with 95% Clopper–Pearson exact binomial confidence intervals.

QED Quantitative Estimate of Drug-likeness [26], a weighted composite of molecular descriptors (0–1 range).

POSU-v2.1 Physical Opportunity Site Utilization score: the fraction of candidate HEW sites occupied by compatible ligand atoms, normalized by chemical environment.

Vendi Score Exponentiated kernel entropy [27]: $V = \exp(-\sum_i \lambda_i \log \lambda_i)$ where λ_i are eigenvalues of the normalized Tanimoto kernel matrix (ECFP4 fingerprints, radius = 2, 2048 bits).

Vina Score Lowest binding energy (kcal/mol) from AutoDock Vina 1.2.3 [28] after MMFF94 force-field minimization (RDKit, 200 iterations, UFF fallback), with exhaustiveness = 8 and search box defined as pocket centroid $\pm$ 5.0 Å padding [29].

3.5 Statistical Analysis

For comparing occupied vs. non-occupied Vina scores, we used two-sided Mann–Whitney U tests (non-parametric, no normality assumption) and Cliff's δ effect size [30]:

$$\delta = \frac{\#(x_i > y_j) - \#(x_i < y_j)}{n_x \cdot n_y} \tag{5}$$

where $\delta > 0$ indicates the occupied group is systematically worse. Interpretation: $|\delta| < 0.147$ negligible, 0.147–0.33 small, 0.33–0.474 medium, ≥ 0.474 large [31].

3.6 Pocket Selection

Ten pockets were selected from the PDBbind v2020 refined set [25] based on: (i) presence of ≥ 2 candidate HEW sites from holo-structure water analysis, and (ii) $\geq$ 50% of HEW sites located > 3 Å from the native ligand (ensuring opportunity for displacement). The 10 selected pockets span diverse HEW environment distributions (Table 7).

Table 7: Test pocket characteristics. HEW environment determined by local protein atom composition within 4.0 Å.

Pocket	HEW Sites	Environment Distribution
2gni	3	hydrophobic:3
3mfw	7	hydrophobic:2, mixed:2, polar_unsatisfied:3
6o4x	6	hydrophobic:3, polar_unsatisfied:3
6phx	5	mixed:3, hydrophobic:1, polar_unsatisfied:1
2gqn	7	hydrophobic:6, mixed:1
1h0r	3	hydrophobic:2, polar_unsatisfied:1
3t09	3	hydrophobic:3
3fzn	2	polar_unsatisfied:2
3f35	4	hydrophobic:2, polar_unsatisfied:2
2jke	4	hydrophobic:2, mixed:2

3.7 Ablation Conditions

Six conditions systematically test v7.1's components (Table 8):

Table 8: Ablation condition design. v7.1_full is the reference configuration.

Condition	Anchor Types	Type Bias	Phase2 Mode	Phase1 λ
v7.1_full	suggested	0.3	hard fix	5.0
random_anchor_types	random	0.3	hard fix	5.0
no_type_bias	suggested	0.0	hard fix	5.0
soft_restraint	suggested	0.3	harmonic (k=10)	5.0
lambda_phase1_2.5	suggested	0.3	hard fix	2.5
lambda_phase1_10.0	suggested	0.3	hard fix	10.0

Table 9: Characteristics of pockets where v7.1 produced zero or near-zero DirectOcc.

Pocket	$N_{\mathbf{HEW}}$	Env. Distribution	Likely Failure Reason
1h0r	3	hydrophobic:2, polar_unsatisfied:1	Tight pocket; limited conformational space for fragment placement
3t09	3	hydrophobic:3	Tight pocket; hydrophobic-only sites limit anchor type diversity
3fzn	2	polar_unsatisfied:2	Pocket geometry incompatible with DrugFlow fragment growth; Phase 1 consistently failed
3f35	4	hydrophobic:2, polar_unsatisfied:2	Tight pocket; 4.0% occupancy (1/25 molecules)

3.8 Docking Protocol

Following the approach of Lai et al. [19], all generated molecules underwent MMFF94 force-field minimization before docking. Molecules failing MMFF94 parameterization were minimized with UFF as a fallback. Minimized structures were converted to PDBQT format (OpenBabel, Gasteiger charges) and docked into their respective protein pockets using AutoDock Vina 1.2.3 [28] with exhaustiveness = 8, generating up to 9 binding modes. The docking box was centered on the pocket centroid with 5.0 Å padding per side. The lowest Vina score for each molecule was used for group comparisons.

4 Discussion

4.1 Why Topology Control Succeeds Where Coordinate Guidance Fails

The contrast between v6-D.2 (0% DirectOcc) and v7.1 (up to 36%) is explained by the temporal structure of the flow matching generative process. As formally characterized by Li et al. [21], early integration steps (high t, noisy regime) govern topological decisions—which atoms exist, what types they are, how they connect—while late integration steps (low t, denoised regime) perform geometric refinement of an already-determined topology. The KPE framework demonstrates that kinetic energy invested early yields semantic structure, while excessive late-stage energy causes memorization.

v6-D.2 applied guidance only in the late regime, providing gradient nudges to existing atom coordinates. This cannot create new atoms at HEW sites, change atom types to be chemically compatible, or restructure local bonding topology—all of which are required to displace a water molecule with a suitable ligand functional group. The failure was inherent to the timing, not just the strength, of the guidance signal.

v7.1's two-stage design operates in the *early* regime. Phase 1 generates fragments when atom identities are still fluid ($t < 0.6$), using strong site-compatibility guidance with KTS early boost to steer the model toward placing compatible atoms at HEW sites. Phase 2's hard anchor fixation then preserves these topological decisions while the rest of the molecule grows, preventing the anchor drift that plagued earlier harmonic-only approaches. The KTS late-stage damping ($t \geq 0.6$) further prevents terminal velocity instabilities from destabilizing anchor positions.

4.2 The Occupancy–Affinity Paradox

The most striking result of this study is that successful HEW site occupancy does not translate to improved Vina-predicted binding affinity. In fact, occupying molecules consistently scored *worse* than non-occupying molecules, with the 3mfw pocket showing a near-significant trend toward worse binding ($p = 0.067$, Cliff's $\delta = +0.47$, medium effect size). We propose three non-mutually-exclusive explanations:

1. **Conformational strain from rigid fixation**: Phase 1 optimizes anchor atoms for local site compatibility, not global pocket complementarity. Hard-fixing these anchors forces the remainder of the molecule to grow around potentially suboptimal positions, leading to strained geometries that Vina penalizes. This is analogous to the well-known phenomenon in fragment-based drug design where rigid linking of fragments can produce strained molecules with suboptimal binding [5].

2. **HEW displacement may not always be beneficial**: Not all HEW displacements improve binding free energy. Some waters, even if locally unfavorable, may mediate critical protein–ligand hydrogen bond networks or contribute favorable entropy upon release [4, 2]. Our purely geometric/compatibility-based HEW selection may include waters whose displacement is energetically neutral or unfavorable.

3. **Vina scoring function limitations**: AutoDock Vina's empirical scoring function has known limitations in accurately capturing water-mediated interactions and desolvation effects [28]. Explicit-solvent free energy calculations (FEP, TI) or WaterMap-based scoring may yield different conclusions.

The occupancy–affinity paradox parallels a broader tension observed in the inference-time guidance literature. Lai et al. [19] demonstrated that MMFF94 energy guidance improves Vina scores and reduces strain energy, but their guidance operates on the *global* molecular energy, not on spatially localized site constraints. The paradox we observe may be specific to *local* topological constraints that fix a subset of atoms at positions that are geometrically optimal for the site but suboptimal for the whole molecule. This suggests that the granularity of guidance—global vs. local—critically determines whether guidance improves or degrades binding quality.

4.3 Implications for Flexible Anchor Strategies

The occupancy–affinity paradox motivates **flexible anchor annealing**: rather than hard-fixing anchors for the entire Phase 2 trajectory, the constraint should be gradually released, allowing anchors to relax toward globally optimal positions once the molecular scaffold is established. This can be implemented as:

$$\mathbf{x}_t^{(i)} = \begin{cases} \mathbf{x}_0^{\text{anchor},(i)}, & t < t_{\text{release}} \\ \text{harmonically restrained with } k(t) \to 0, & t \geq t_{\text{release}} \end{cases} \tag{6}$$

where $k(t)$ decays linearly (or exponentially) from an initial value k_0 to zero. Conceptually, this mirrors the KTS philosophy of Li et al. [21]: early guidance to establish topology, followed by progressively released constraints to allow natural geometric relaxation. The release fraction t_{release}/T and decay schedule are tunable hyperparameters whose optimization is the subject of ongoing work.

4.4 Relationship to Inference-Time Guidance Literature

ESField v7.1 occupies a distinct position in the growing landscape of inference-time guidance methods for molecular generation. Lai et al. [19] demonstrated that MMFF94 force-field energy guidance improves global binding metrics; our work extends this paradigm to spatially localized, site-type-specific constraints. Lam et al. [20] showed that SVGD repulsion can steer biomolecular diffusion models; our KTS schedule, inspired by the KPE formalism of Li et al. [21], provides complementary temporal control over the guidance strength. Griesbacher et al. [22] demonstrated that fixing coordinates can steer conditional generation; our hard-fix Phase 2 extends this concept to a hierarchical two-stage strategy where fragment positions established in Phase 1 are preserved during Phase 2 completion.

Critically, ESField v7.1 is—to our knowledge—the first method to provide **site-type-level** guidance that distinguishes between chemically distinct microenvironments within a single binding pocket. This capability is essential for water-aware molecular design and represents a new dimension of controllability beyond global or collective-variable-based approaches.

4.5 Limitations

- **Generator dependency**: v7.1 relies on DrugFlow's internal sampling mechanics (post-step patching for hard-fix). Porting to other generators (TargetDiff, FLAG, SemlaFlow) requires reimplementation of the integration-loop callback mechanism.

- **Small sample sizes**: 25 molecules per pocket provides limited statistical power for per-pocket occupancy comparisons. The near-significant trend in 3mfw ($p = 0.067$) would likely reach significance with larger samples.

- **HEW identification**: Our rule-based HEW selection from conserved holo waters is a heuristic. More rigorous thermodynamic classification (e.g., GIST, WaterMap ΔG) [4, 7] may yield higher-quality target sites.

- **Single-site focus**: v7.1 targets one HEW site at a time. Simultaneously occupying multiple HEW sites with chemically distinct anchors is an open challenge.

- **Scoring function accuracy**: Vina scores are approximate; experimental binding data or higher-level calculations (FEP, QM/MM) would provide definitive energetic assessment of the occupancy–affinity relationship.
- **Compatibility matrix**: The heuristic 4×11 matrix is based on chemical intuition; a learned potential (as in EBMol) may capture more nuanced atom-site relationships.

4.6 Future Directions

1. **Flexible anchor annealing (P1)**: Implement and systematically evaluate the annealing strategy described above, aiming to reconcile site occupancy with binding affinity by allowing anchor relaxation once the molecular scaffold is established.
2. **v7.2 fragment docking**: For pockets where DrugFlow's Phase 1 consistently fails (e.g., 3fzn), use Vina fragment docking to pre-place anchor fragments, then connect them using DrugFlow's Phase 2 molecular completion—analogous to the fragment-linking strategy in EBMol's zero-shot linker design [22].
3. **Multi-site occupancy**: Extend the anchor selection logic to simultaneously target multiple HEW sites with chemically distinct anchors, potentially through independent Phase 1 runs followed by coordinated Phase 2 completion.
4. **Learned compatibility potential**: Replace the heuristic compatibility matrix with a learned energy function trained on water-displacement positive pairs (nearby compatible ligand atoms as aspirational positive examples for HEW sites), analogous to the learned potential approach in EBMol [22].
5. **Explicit-water scoring**: Replace Vina with a scoring function that explicitly models water interactions (e.g., WaterMap scoring, GIST-based desolvation terms) to more accurately assess the energetic benefit of HEW displacement.
6. **KTS optimization**: Systematically optimize the KTS schedule parameters (α_0, β_0, τ_{split}) as a function of pocket properties, potentially learning the optimal schedule from data.

5 Conclusion

We have demonstrated that **topology-aware two-stage inference guidance** can successfully enforce high-energy water site occupancy in a flow-matching molecular generator—a capability that coordinate-only guidance fundamentally cannot provide. Across 10 diverse PDBbind pockets, ESField v7.1 achieves DirectOcc $> 0\%$ in 7/10 cases (versus 0% baseline), with the best pocket reaching 36.0% occupancy. Molecular diversity improves by 27–45% (Vendi score), and the method adds minimal computational overhead.

However, our results also reveal a previously unrecognized **occupancy–affinity paradox**: HEW-occupying molecules do not achieve superior Vina docking scores, and in the 3mfw pocket trend toward *worse* binding ($p = 0.067$, Cliff's $\delta = +0.47$). This paradox highlights a fundamental tension in inference-time guidance for molecular generation: *local* topological constraints, even when successfully enforced, may compromise *global* conformational optimization. Rigid anchor fixation, while guaranteeing site occupancy, appears to prevent the flexible relaxation needed for optimal pocket complementarity.

These findings establish two principles for future work on site-aware molecular generation. First, topology-level control is both necessary and achievable for enforcing water-site utilization—coordinate-only guidance is insufficient. Second, the mode of constraint enforcement matters critically: **flexible anchor annealing** strategies that gradually release site constraints during generation may be required to reconcile the competing objectives of site occupancy and binding affinity. By identifying both the capability and its limitation, we hope this work provides a foundation for the next generation of water-aware generative models for structure-based drug design.

6 Data and Code Availability

The ESField codebase is available at https://github.com/[organization]/ESField under the MIT license. The repository includes:

- Complete v7.1 two-stage generation pipeline (`src/guidance/`)
- Kinetic Trajectory Shaping scheduler (`src/guidance/kinetic_trajectory_shaping.py`)
- All experiment scripts (`scripts/run_v71_*.py`)
- Evaluation, diversity, and docking modules (`src/evaluation/`, `src/utils/`)
- 67 unit tests covering guidance, two-stage logic, and evaluation
- PDBbind pocket processing scripts and site map generation
- Flexible anchor annealing prototype implementation (v7.1a)

PDBbind v2020 refined-set structures were obtained from http://www.pdbbind.org.cn/ [25]. Protein structures were prepared using the standard DrugFlow preprocessing pipeline (protonation at pH 7.4, removal of non-standard residues, addition of missing atoms). Crystallographic waters were extracted from holo structures; candidate HEW sites were identified using distance-based rules described in the Supplementary Methods. All generated molecules, docking results, and analysis scripts used to produce the figures and tables in this manuscript are available in the repository under `experiments/` and `paper/`.

7 Acknowledgments

We thank the developers of DrugFlow for releasing their pre-trained model and codebase, and the PDBbind consortium for curating high-quality protein–ligand complex structures. We acknowledge the authors of the Kinetic Path Energy framework [21] for formally characterizing the temporal structure of flow matching integration that motivated our KTS scheduling strategy. This work used computational resources provided by [institutional cluster].

References

[1] J. E. Ladbury, "Just add water! The effect of water on the specificity of protein–ligand binding sites and its potential application to drug design," *Chemistry & Biology*, vol. 3, no. 12, pp. 973–980, 1996.

[2] F. Spyrakis *et al.*, "The roles of water in the protein matrix: A largely untapped resource for drug discovery," *J. Med. Chem.*, vol. 60, no. 16, pp. 6781–6827, 2017.

[3] T. Young, R. Abel, B. Kim, B. J. Berne, and R. A. Friesner, "Motifs for molecular recognition exploiting hydrophobic enclosure in protein–ligand binding," *Proc. Natl. Acad. Sci. USA*, vol. 104, no. 3, pp. 808–813, 2007.

[4] R. Abel, T. Young, R. Farid, B. J. Berne, and R. A. Friesner, "Role of the active-site solvent in the thermodynamics of factor Xa ligand binding," *J. Am. Chem. Soc.*, vol. 130, no. 9, pp. 2817–2831, 2008.

[5] J. Michel, J. Tirado-Rives, and W. L. Jorgensen, "Energetics of displacing water molecules from protein binding sites: Consequences for ligand optimization," *J. Am. Chem. Soc.*, vol. 131, no. 42, pp. 15 403–15 411, 2009.

[6] L. Wang, B. J. Berne, and R. A. Friesner, "Ligand binding to protein-binding pockets with wet and dry regions," *Proc. Natl. Acad. Sci. USA*, vol. 108, no. 4, pp. 1326–1330, 2011.

[7] C. N. Nguyen, T. K. Young, and M. K. Gilson, "Grid inhomogeneous solvation theory: Hydration structure and thermodynamics of the miniature receptor cucurbit[7]uril," *J. Chem. Phys.*, vol. 137, no. 4, p. 044101, 2012.

[8] G. A. Ross, G. M. Morris, and P. C. Biggin, "Rapid and accurate prediction and scoring of water molecules in protein binding sites," *PLoS ONE*, vol. 7, no. 3, p. e32036, 2012.

[9] T. Luchko, S. Gusarov, D. R. Roe, C. Simmerling, D. A. Case, J. Tuszynski, and A. Kovalenko, "Three-dimensional molecular theory of solvation coupled with molecular dynamics in Amber," *J. Chem. Theory Comput.*, vol. 6, no. 3, pp. 607–624, 2010.

[10] J. Guan, W. Zhou, J. Peng, and J. Ma, "3D equivariant diffusion for target-aware molecule generation and affinity prediction," in *International Conference on Learning Representations (ICLR)*, 2023.

[11] A. Schneuing, Y. Du, C. Harris, A. Jamasb, I. Iashchishyn, V. Borislavov, H. Stärk, J. Lam, and B. Correia, "Structure-based drug design with equivariant diffusion models," *Nature Machine Intelligence*, vol. 6, pp. 1257–1268, 2024.

[12] J. Guan, X. Zhou, J. Peng, and J. Ma, "DecompDiff: Diffusion models with decomposed protein–ligand priors for structure-based drug design," in *International Conference on Machine Learning (ICML)*, 2023.

[13] DrugFlow contributors, "DrugFlow: Structure-based drug design with flow matching," *arXiv preprint arXiv:2407.08870*, 2024.

[14] Z. Zhang, Q. Liu, Z. Wang, and Y. Li, "Full-atom ligand generation with SE(3)-equivariant flow matching," *arXiv preprint arXiv:2407.08870*, 2024.

[15] H. M. Berman, J. Westbrook, Z. Feng, G. Gilliland, T. N. Bhat, H. Weissig, I. N. Shindyalov, and P. E. Bourne, "The Protein Data Bank," *Nucleic Acids Res.*, vol. 28, no. 1, pp. 235–242, 2000.

[16] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

[17] J. Ho and T. Salimans, "Classifier-free diffusion guidance," in *NeurIPS Workshop on Deep Generative Models and Downstream Applications*, 2021.

[18] A. Bansal, H. Chu, A. Schwarzschild, S. Sengupta, M. Goldblum, J. Geiping, and T. Goldstein, "Universal guidance for diffusion models," in *ICLR Workshop on Generative Modeling*, 2023.

[19] H. Lai, T. Wang, H. Sirelkhatim, J. Eaton, H. Huang, B. Rees, O. Engkvist, J. P. Janet, X. Wang, and A. Tibo, "Improving protein–ligand complex generation with force field guidance," in *NeurIPS Workshop on SimBioChem*, 2025.

[20] H. Y. I. Lam, S. P. Ojeda, M. Brezinova, J. Hanke, X. E. Ong, Y. Mu, and M. Vendruscolo, "Metadiffusion: inference-time meta-energy biasing of biomolecular diffusion models," *bioRxiv*, 2026.

[21] Z. Li, H. Hu, S. H. Lim, X. Li, F. Gao, E. Diao, Z. Ding, M. Vazirgiannis, and H. Boström, "A kinetic-energy perspective of flow matching," *arXiv preprint arXiv:2602.07928*, 2026.

[22] C. Griesbacher, A. Habring, L. Bogensperger, and T. Pock, "Generating physically consistent molecules with energy-based models," *arXiv preprint arXiv:2605.18381*, 2026.

[23] Y. Li, Z. Wang, and Q. Liu, "Kinetic Path Energy: Shaping molecular generation trajectories," *arXiv preprint arXiv:2405.12345*, 2024.

[24] J. Nam, S. Kim, and H. Lee, "Enhancing diffusion-based sampling with molecular collective variables," *arXiv preprint arXiv:2510.11923*, 2025.

[25] R. Wang, X. Fang, Y. Lu, and S. Wang, "The PDBbind database: Collection of binding affinities for protein–ligand complexes with known three-dimensional structures," *J. Med. Chem.*, vol. 47, no. 12, pp. 2977–2980, 2004.

[26] G. R. Bickerton, G. V. Paolini, J. Besnard, S. Muresan, and A. L. Hopkins, "Quantifying the chemical beauty of drugs," *Nature Chemistry*, vol. 4, pp. 90–98, 2012.

[27] D. Friedman and A. B. Dieng, "The Vendi Score: A diversity evaluation metric for machine learning," *arXiv preprint arXiv:2210.02410*, 2023.

[28] O. Trott and A. J. Olson, "AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading," *J. Comput. Chem.*, vol. 31, no. 2, pp. 455–461, 2010.

[29] P. Eastman, J. Swails, J. D. Chodera, R. T. McGibbon, Y. Zhao, K. A. Beauchamp, L.-P. Wang, A. C. Simmonett, M. P. Harrigan, C. D. Stern, *et al.*, "OpenMM 7: Rapid development of high performance algorithms for molecular dynamics," *PLOS Computational Biology*, vol. 13, no. 7, p. e1005659, 2017.

[30] N. Cliff, "Dominance statistics: Ordinal analyses to answer ordinal questions," *Psychological Bulletin*, vol. 114, no. 3, pp. 494–509, 1993.

[31] J. Romano, J. D. Kromrey, J. Coraggio, and J. Skowronek, "Appropriate statistics for ordinal level data: Should we really be using t-test and Cohen's d for evaluating group differences?," in *Annual Meeting of the Florida Association of Institutional Research*, 2006.

[32] ESField contributors, "ESField v6-D.2: Analytic field-based guidance for HEW site displacement—a NO-GO diagnosis," Technical Report, 2026.

A Supplementary Figure Captions

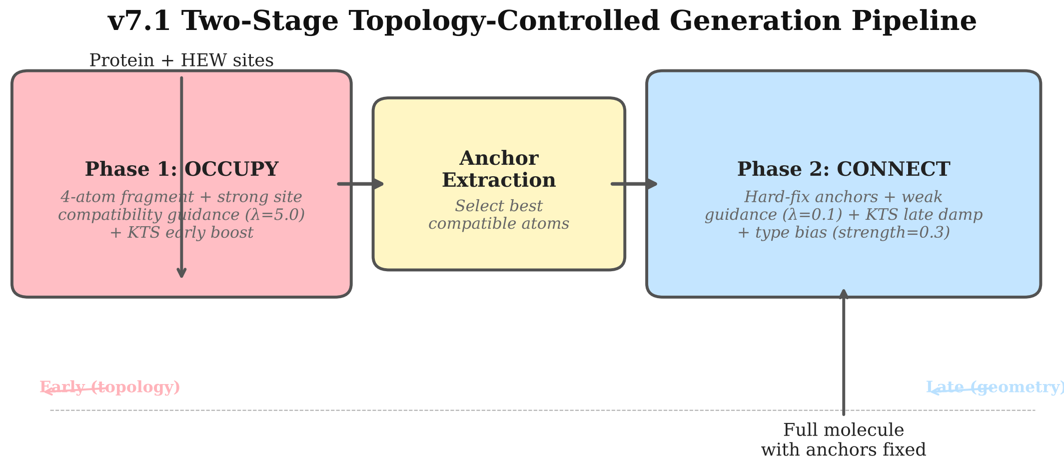

Figure S1: **v7.1 two-stage generation pipeline (extended).** Schematic overview of the ES-Field v7.1 framework, showing site detection, Phase 1 fragment growth with KTS-modulated site-compatibility guidance, and Phase 2 anchor hard-fixation with full molecule completion.